

INTRODUCTION TO FLUID MECHANICS

QUIZ (Sep 30, 2013)

Given the steady planar motion described by the velocity components

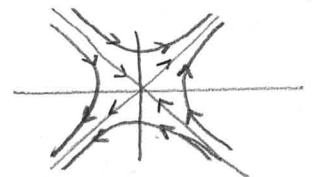
$$v_x = Ae^{\Omega t}y \quad \text{and} \quad v_y = Be^{\Omega t}x,$$

where A , B , and Ω are constants.

1. Determine the value of the volume dilation rate $\nabla \cdot \vec{v}$.
2. Obtain the value of B for which the resulting flow is irrotational. Adopt this particular value in what follows.
3. Calculate the stream line that intersects the point $(x, y) = (x_0, y_0)$ at a given instant of time t . Sketch the resulting stream lines.
4. Compute the trajectory of a fluid particle located initially at $(x, y) = (x_0, y_0)$.
5. If they exist, calculate the velocity potential φ and the stream function ψ .
6. Determine the rate-of-strain tensor $\bar{T}_d$.
7. Describe the deformation of a fluid element of square shape whose sides of length dl are initially oriented parallel to the axes.

$$1) \quad \boxed{\nabla \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0} \quad \left| \quad 2) \quad \nabla \wedge \vec{v} = \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \vec{e}_z = Be^{\Omega t} - Ae^{\Omega t} = 0 \Rightarrow \boxed{B=A} \right.$$

$$3) \quad \frac{dx}{v_x} = \frac{dy}{v_y} \Rightarrow \frac{dx}{y} = \frac{dy}{x} \rightarrow \boxed{y^2 - x^2 = y_0^2 - x_0^2}$$



$$4) \quad \left. \begin{aligned} \frac{dx}{dt} &= Ae^{\Omega t}y \\ \frac{dy}{dt} &= Ae^{\Omega t}x \end{aligned} \right\} \begin{aligned} t=0: \\ x=x_0 \\ y=y_0 \end{aligned} \quad \frac{dx}{dy} = \frac{y}{x} \rightarrow \boxed{y^2 - x^2 = y_0^2 - x_0^2}$$

PATH LINES $\equiv$ STREAM LINES

$$\int_{x_0}^x \frac{dx}{\sqrt{x^2 + y_0^2 - x_0^2}} = \pm \int_0^t Ae^{\Omega t} dt \Rightarrow \ln \left(\frac{x + \sqrt{x^2 + y_0^2 - x_0^2}}{x_0 + y_0} \right) = \pm \frac{A}{\Omega} (e^{\Omega t} - 1)$$

$$5) \quad \left. \begin{aligned} v_x &= \frac{\partial \varphi}{\partial x} = Ae^{\Omega t}y \\ v_y &= \frac{\partial \varphi}{\partial y} = Ae^{\Omega t}x \end{aligned} \right\} \Rightarrow \boxed{\varphi = Ae^{\Omega t}xy + \varphi_0}$$

$$\left. \begin{aligned} v_x &= \frac{\partial \psi}{\partial y} = Ae^{\Omega t}y \\ -v_y &= \frac{\partial \psi}{\partial x} = -Ae^{\Omega t}x \end{aligned} \right\} \Rightarrow \boxed{\psi = Ae^{\Omega t}(y^2 - x^2) + \psi_0}$$

$$6) \quad \bar{T}_d = \frac{1}{2} (\nabla \vec{v} + \nabla \vec{v}^T) = \begin{bmatrix} 0 & Ae^{\Omega t} \\ Ae^{\Omega t} & 0 \end{bmatrix} \quad 7)$$

